Name

Mathematics Stage 5 Path

Coordinate Geometry Formulas

Book 1 Apply the midpoint, gradient, and distance formulas

Version: 261007

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Syllabus Content

MA5-LIN-P-01 describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems

Apply formulas to find the midpoint and gradient/slope of an interval on the Cartesian plane

- Apply the formula to find the midpoint of the interval joining 2 points on the Cartesian plane: $M(x,y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
- Use the relationship $m = \frac{\text{rise}}{\text{run}}$ to establish the formula for the gradient/slope (m) of the interval joining the 2 points $(x_1,\ y_1)$ and (x_2, y_2) on the Cartesian plane: $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Apply the gradient formula to find the gradient of the interval joining 2 points on the Cartesian plane

Apply the distance formula to find the distance between 2 points located on the Cartesian plane

- Apply knowledge of Pythagoras' theorem to establish the formula for the distance (d) between the 2 points $(x_1,\ y_1)$ and (x_2, y_2) on the Cartesian plane: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Apply the distance formula to find the distance between 2 points on the Cartesian plane

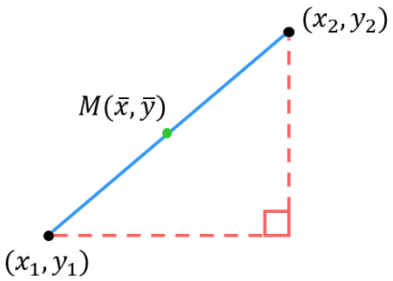
Midpoint Formula

Midpoint Formula

For two coordinates (x_1, y_1) and (x_2, y_2):

$$M = (\text{mean of } x, \text{Mean of y})$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$



Example Calculate midpoint using the formula

Calculate the midpoint of the interval between these points.

$(-4, 3)$ and $(8, -2)$

x_1 y_1 x_2 y_2

$$\bar{x} = \frac{x_1 + x_2}{2} \qquad \bar{y} = \frac{y_1 + y_2}{2}$$

$$= \frac{-4 + 8}{2} = 2 \qquad = \frac{3 + (-2)}{2} = 0.5$$

$M = (2, 0.5)$

$(-3, -5)$ and $(2, 8)$

x_1 y_1 x_2 y_2

$$\bar{x} = \frac{x_1 + x_2}{2} \qquad \bar{y} = \frac{y_1 + y_2}{2}$$

$$= \frac{-3 + 2}{2} = -0.5 \qquad = \frac{-5 + 8}{2} = 1.5$$

$M = (-0.5, 1.5)$

Guided Practice

Calculate the midpoint of the interval between these points.

a $(2,3)$ and $(6,9)$

x_1 y_1 x_2 y_2

$$\bar{x} = \frac{x_1 + x_2}{2} \qquad \bar{y} = \frac{y_1 + y_2}{2}$$

$M = (\quad , \quad)$

$(4, 6)$

b $(-2,6)$ and $(8, -1)$

$$\bar{x} = \frac{x_1 + x_2}{2} \qquad \bar{y} = \frac{y_1 + y_2}{2}$$

$M = (\quad , \quad)$

$(3, 2.5)$

c $(-5.5, 4.5)$ and $(7, -1)$

$(0.75, 1.75)$

d $(-5, -6)$ and $(2,4)$

$(-1.5, -1)$

Key Ideas

1. The midpoint is the …… of the x and y coordinates.

1 Find the midpoint of the x coordinates and the find the midpoint of the y coordinates.

a $(2,7)$ and $(14,15)$

$\bar{x} = \frac{\square + \square}{2} = \square$

$\bar{y} = \frac{\square + \square}{2} = \square$

$M(\bar{x}, \bar{y}) = (\square, \square)$

$(8, 11)$

b $(5,2)$ and $(23,14)$

$\bar{x} = \frac{\square + \square}{2} = \square$

$\bar{y} = \frac{\square + \square}{2} = \square$

$M(\bar{x}, \bar{y}) = (\square, \square)$

$(14, 8)$

c $(1,14)$ and $(19,22)$

$\bar{x} = \frac{\square + \square}{2} = \square$

$\bar{y} = \frac{\square + \square}{2} = \square$

$M(\bar{x}, \bar{y}) = (\square, \square)$

$(10, 18)$

d $(14,2)$ and $(8,6)$

$\bar{x} =$ .....

$\bar{y} =$ .....

$M = (\quad , \quad)$

$(11, 4)$

e $(9,17)$ and $(3,8)$

$\bar{x} =$ .....

$\bar{y} =$ .....

$M = (\quad , \quad)$

$(6, 12.5)$

f $(6,0)$ and $(11,13)$

$\bar{x} =$ .....

$\bar{y} =$ .....

$M = (\quad , \quad)$

$(8.5, 6.5)$

g $(-4, -3)$ and $(8, -11)$

$(2, -7)$

h $(2, -3)$ and $(-6,9)$

$(-2, 3)$

i $(-1, -6)$ and $(-3,10)$

$(-2, 2)$

j $(7, -5)$ and $(-3,9)$

$(2, 2)$

k $(-9, -2)$ and $(-1, -9)$

$(-5, -5.5)$

l $(0, -5)$ and $(-10,4)$

$(-5, -0.5)$

2 Calculate the midpoint of these coordinates.

a $(0,0)$ and $(4,2)$

(2, 1)

b $(0,0)$ and $(2,4)$

(1, 2)

c $(0,0)$ and $(4,8)$

(2, 4)

d $(4,8)$ and $(0,0)$

(2, 4)

e $(4,8)$ and $(2,2)$

(3, 5)

f $(4,8)$ and $(4,4)$

(4, 6)

g $(4,8)$ and $(3,3)$

(3.5, 5.5)

h $(4,8)$ and $(9,9)$

(6.5, 8.5)

i $(4,8)$ and $(10,10)$

(7, 9)

j $(4,8)$ and $(1,1)$

(2.5, 4.5)

k $(4,8)$ and $(-1,1)$

(1.5, 4.5)

l $(4,8)$ and $(-1,-1)$

(1.5, 3.5)

m $(-4,8)$ and $(-1,-1)$

(-2.5, 3.5)

n $(4,8)$ and $(-2,-2)$

(1, 3)

o $(-4,8)$ and $(-2,-2)$

(-3, 3)

p $(-4,-8)$ and $(-2,-2)$

(-3, -5)

q $(-2,-4)$ and $(-2,-2)$

(-2, -3)

r $(-20,-40)$ and $(-20,-20)$

(-20, -30)

s $(-200,-400)$ and $(-200,-200)$

(-200, -300)

t $(-200,-200)$ and $(200,200)$

(0, 0)

3 Calculate the midpoint of these coordinates.

a $(-2,-2)$ and $(6,-3)$

(2, -2.5)

b $(0,6)$ and $(6,0)$

(3, 3)

c $(2.5,3)$ and $(4,-7)$

(3.25, -2)

d $(0.5,0)$ and $(5,3)$

(2.75, 1.5)

e $(51,-12)$ and $(-36,11)$

(7.5, -0.5)

f $(2a,4b)$ and $(4a,6b)$

(3a, 5b)

g $(5p,3q)$ and $(-3p,5q)$

(p, 4q)

h $(-4r,7s)$ and $(4r,-7s)$

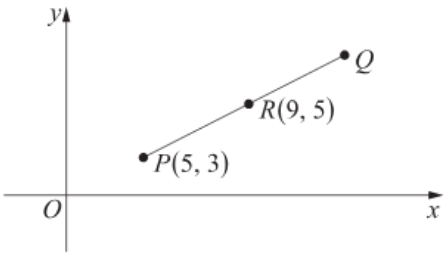
(0, 0)

4 Find the missing coordinates if M is the midpoint of A and B

A	B	M
$(4,2)$	(8, 0)	$(6,1)$
$(7,1)$	(-3, 1)	$(2,1)$
(1, -4)	$(3,6)$	$(2,1)$
(-6, 5)	$(0,-1)$	$(-3,2)$
(-6, 9)	$(4,4)$	$(-1,6.5)$

5 **2018 HSC Advanced Band 2**

The point $(9,5)$ is the midpoint of the interval PQ, where P has coordinates $(5,3)$.
What are the coordinates of Q?



NOT TO SCALE

(13, 7)

6 Find the values of a and b when:

a midpoint of $(a,3)$ and $(7,b)$ is $(5,4)$.

$\bar{x} = \frac{x_1 + x_2}{2}$ $\bar{y} = \frac{y_1 + y_2}{2}$

$5 = \frac{a+7}{2}$

$a = 3, b = 5$

b midpoint of $(a,-1)$ and $(2,b)$ is $(-1,2)$.

$a = -4, b = 5$

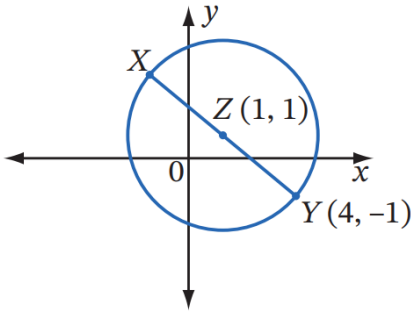
c midpoint of $(-3,a)$ and $(b,2)$ is $\left(-\frac{1}{2},0\right)$.

$a = -2, b = 2$

d midpoint of $(-5,a)$ and $(b,-4)$ is $\left(-\frac{3}{2},\frac{7}{2}\right)$.

$a = 11, b = 2$

7 The circle has diameter XY and centre Z. What are the coordinates of X?



Midpoint of diameter gives centre:

$\frac{x_X + 4}{2} = 1$ and $\frac{y_X + (-1)}{2} = 1$

$x_X + 4 = 2$

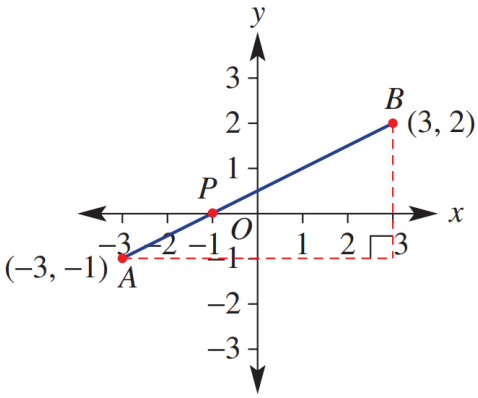
$x_X = -2$

$y_X - 1 = 2$

$y_X = 3$

$X(-2, 3)$

8 From left to right, this line segment shows point $P(-1, 0)$ dividing the segment in the ratio 1 : 2.



a What fraction of the horizontal distance between the end points is P from A?

$\frac{1}{3}$

b What fraction of the vertical distance between the end points is P from A?

$\frac{1}{3}$

c Hence find P algebraically.

Coordinates of P are one third the distance from A to B

x coordinate: $-3 + \frac{1}{3}(3 - -3) = -1$

y coordinate: $-1 + \frac{1}{3}(2 - -1) = 0$

$(-1, 0)$

9 Find the coordinates of point P that divides $A(-4, 9)$ and $B(1, -1)$ in the ratio 2 : 3.

Coordinates of P are two fifths the distance from A to B

x coordinate: $-4 + \frac{2}{5}(1 - (-4)) = -2$

y coordinate: $9 + \frac{2}{5}(-1 - 9) = = 5$

$P(-2, 5)$

10 Find the coordinates of point P that divides $A(-2, -3)$ and $B(4, 0)$ in the ratio 3 : 2.

Coordinates of P are three fifths the distance from A to B

x coordinate: $-2 + \frac{3}{5}(4 - (-2)) = \frac{8}{5}$

y coordinate: $-3 + \frac{3}{5}(0 - (-3)) = -\frac{6}{5}$

$P\left(\frac{8}{5}, -\frac{6}{5}\right)$

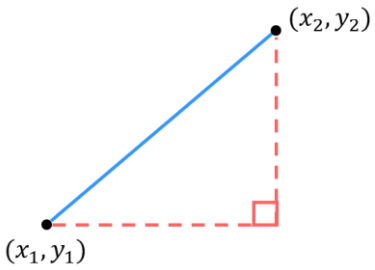
Gradient Formula

Gradient Formula

For two coordinates (x_1, y_1) and (x_2, y_2):

$$m = \frac{\text{rise}}{\text{run}}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Example Calculate gradient using the formula

Calculate the gradient of the interval between these points.

$(4, 2)$ and $(12, 4)$

$x_1\ y_1 \qquad x_2\ y_2$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{4 - 2}{12 - 4}$$

$$= \frac{2}{8} = \frac{1}{4}$$

$(-3, 8)$ and $(5, -2)$

$x_1\ y_1 \qquad x_2\ y_2$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-2 - 8}{5 - (-3)}$$

$$= -\frac{10}{8} = -\frac{5}{4}$$

Guided Practice

Calculate the gradient of the interval between these points.

a $(2,3)$ and $(5,8)$

$x_1\ y_1 \qquad x_2\ y_2$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$=$

$=$ $\frac{5}{3}$

b $(-2,5)$ and $(0, -6)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$=$

$=$ $-\frac{11}{2}$

c $(-1, -2)$ and $(-3, -7)$

$\frac{5}{2}$

d $(4, -3)$ and $(-4,1)$

$-\frac{1}{2}$

Key Ideas

1. The gradient is the change in …… divided by the change in ……

1 Find the gradient of each interval.

a $A(1,4), B(5,0)$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$=$

$=$

-1

b $A(-2,-7), B(3,3)$

2

c $A(-5,-2), B(3,2)$

$\frac{1}{2}$

d $A(3,6), B(5,5)$

$-\frac{1}{2}$

e $A(-1,-2), B(1,4)$

3

f $A(-5,7), B(15,-7)$

$-\frac{7}{10}$

2 A line joins the two points (−1, 3) and (4, −2).

a Calculate the gradient of the line, using $(x_1, y_1) = (-1, 3)$ and $(x_2, y_2) = (4, -2)$.

$m = \frac{-2 - 3}{4 - (-1)} = -\frac{5}{5} = -1$

b Calculate the gradient of the line, using $(x_1, y_1) = (4, -2)$ and $(x_2, y_2) = (-1, 3)$.

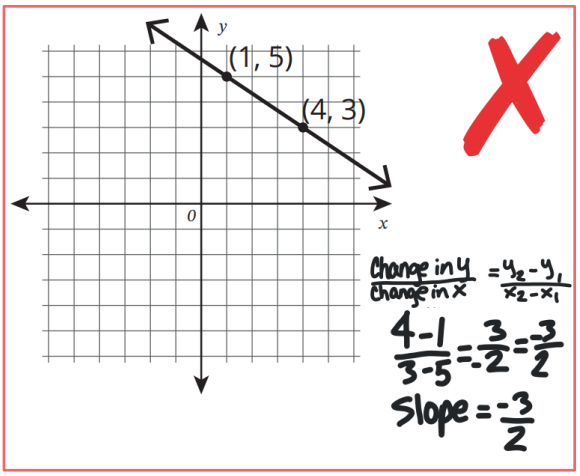
$m = \frac{3 - (-2)}{-1 - 4} = \frac{5}{-5} = -1$

c Do you get the same gradient even if you switch (x_1, y_1) and (x_2, y_2)?

Yes

3 Read the example carefully and answer the questions.

Akemi's work:
Calculate the gradient.



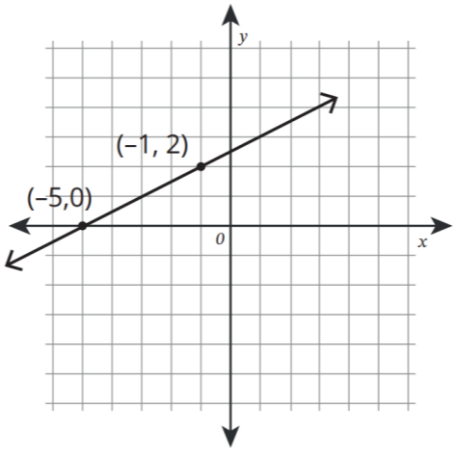
What did Akemi do wrong when substituting the points into the slope formula?

She did change in x over change in y

Using the graph, how could she have checked her work?

Use the gridlines to see if the slope goes 3 down for every 2 across.

Calculate the gradient:



$\frac{1}{2}$

4 **Collinear** points lie on the same line. Three points A, B, C are collinear if AB and BC have the same gradient, because then AB and BC are parallel lines passing through a common point B.

a Determine whether $A(-2,5)$, $B(1,3)$ and $C(7,-1)$ are collinear.

$m_{AB} =$

$m_{BC} =$

collinear

b $P(-2,7), Q(1,1), R(4,-6)$

not collinear

c $P(-5,-4), Q(-2,-2), R(1,0)$

collinear

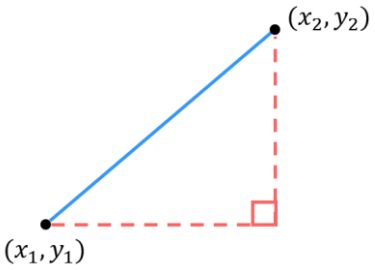
Distance Formula

Distance Formula

For two coordinates (x_1, y_1) and (x_2, y_2):

$$d = \sqrt{\text{run}^2 + \text{rise}^2}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Example Calculate distance using the formula

Calculate the distance of the interval through these points. Answer as a surd and to 2 d.p.

$(2,6)$ and $(5,1)$

x_1 y_1 x_2 y_2

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(5-2)^2 + (1-6)^2}$$
$$= \sqrt{34}$$
$$\approx 5.83$$

$(-3,5)$ and $(6,-4)$

x_1 y_1 x_2 y_2

$$d = \sqrt{(6-(-3))^2 + (-4-5)^2}$$
$$= \sqrt{(9)^2 + (-9)^2}$$
$$= \sqrt{162}$$
$$\approx 12.73$$

Guided Practice

Calculate the length of the interval between these points. Answer as a surd and to 2 d.p.

a $(-2,3)$ and $(5,4)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(\quad)^2 + (\quad)^2}$$
$$= \sqrt{\quad}$$

$5\sqrt{2}$ or 7.07

b $(0,4)$ and $(3,6)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(\quad)^2 + (\quad)^2}$$
$$= \sqrt{\quad}$$

$\sqrt{13}$ or 3.61

c $(-3,4)$ and $(2,4)$

5

d $(1,-2)$ and $(-4,3)$

$5\sqrt{2}$ or 7.07

Key Ideas

1. We calculate the distance between two points using ……'s theorem.

1 Find the length, or distance, of each interval. Answer to 1 d.p. where necessary.

a $A(1,4), B(5,1)$

5

b $A(-2,7), B(3,-5)$

13

c $A(-5,-2), B(3,4)$

10

d $A(3,6), B(5,4)$

2.8

e $A(-4,-1), B(4,3)$

8.9

f $A(5,-12), B(0,0)$

13

2 Find the length between these intervals without drawing it. Answer in exact form.

a $(0,0)$ and $(6,8)$

10

b $(0,0)$ and $(-6,8)$

10

c $(0,0)$ and $(-6,-8)$

10

d $(0,0)$ and $(-6,-8)$

10

e $(1,1)$ and $(7,9)$

10

f $(1,1)$ and $(8,10)$

$\sqrt{130}$

g $(1,1)$ and $(-8,10)$

$\sqrt{162}$

h $(1,1)$ and $(8,-10)$

$\sqrt{170}$

i $(1,1)$ and $(-1,-1)$

$\sqrt{8}$

j $(15,1)$ and $(8,12)$

$\sqrt{170}$

k $(15,23)$ and $(8,12)$

$\sqrt{170}$

l $(1,1)$ and $(-16,20)$

$\sqrt{650}$

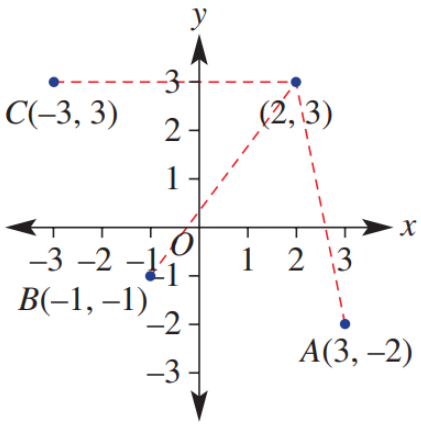
m $(1,1)$ and $(-17,19)$

$\sqrt{648}$

n $(-8,10)$ and $(-17,19)$

$\sqrt{162}$

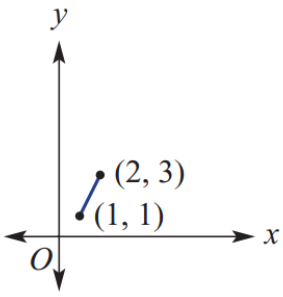
3 Which of the points $A, B,$ or C are closest to the point $(2,3)$?



B and C are both 5 units away from (2, 3)

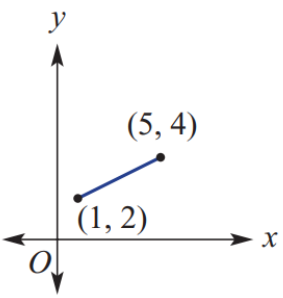
4 Calculate the length, gradient, and midpoint of each line segment AB.
Give your answers to 1 decimal place where necessary.

a



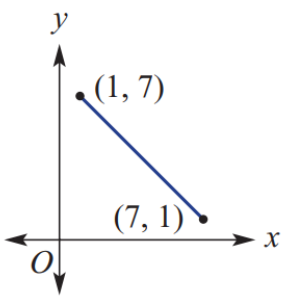
Length: $\sqrt{(2-1)^2 + (3-1)^2} = 2.2$
Gradient: $\frac{3-1}{2-1} = 2$
Midpoint: $\left(\frac{1+2}{2}, \frac{1+3}{2}\right) = (1.5, 2)$

b



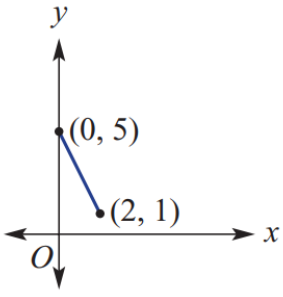
Length: $\sqrt{(5-1)^2 + (4-2)^2} = 4.5$
Gradient: $\frac{4-2}{5-1} = 0.5$
Midpoint: $\left(\frac{1+5}{2}, \frac{2+4}{2}\right) = (3, 3)$

c



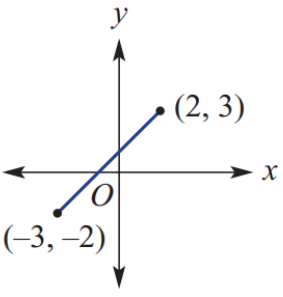
Length: $\sqrt{(7-1)^2 + (1-7)^2} = 8.5$
Gradient: $\frac{1-7}{7-1} = -1$
Midpoint: $\left(\frac{1+7}{2}, \frac{7+1}{2}\right) = (4, 4)$

d



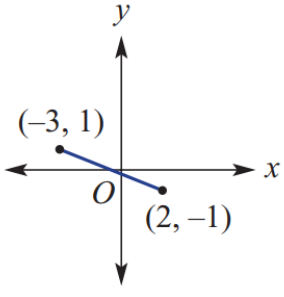
Length: $\sqrt{(2-0)^2 + (1-5)^2} = 4.5$
Gradient: $\frac{1-5}{2-0} = -2.0$
Midpoint: $\left(\frac{0+2}{2}, \frac{5+1}{2}\right) = (1.0, 3.0)$

e



Length: $\sqrt{(2-(-3))^2 + (3-(-2))^2} = 7.1$
Gradient: $\frac{3-(-2)}{2-(-3)} = 1$
Midpoint: $\left(\frac{-3+2}{2}, \frac{-2+3}{2}\right) = (-0.5, 0.5)$

f



Length: $\sqrt{(2-(-3))^2 + (-1-1)^2} = 5.4$
Gradient: $\frac{-1-1}{2-(-3)} = -0.4$
Midpoint: $\left(\frac{-3+2}{2}, \frac{1+(-1)}{2}\right) = (-0.5, 0)$

5 Find the value of a when the distance between:

a $(1, a)$ and $(3,5)$ is $\sqrt{8}$

$$\sqrt{(3-1)^2+(5-a)^2} = \sqrt{8}$$
$$4+(5-a)^2 = 8$$
$$(5-a)^2 = 4$$
$$5-a = \pm 2$$
$$a = 3 \text{ or } 7$$

b $(a, -1)$ and $(4, -3)$ is $\sqrt{29}$

$$\sqrt{(4-a)^2+\left(-3-(-1)\right)^2} = \sqrt{29}$$
$$(4-a)^2+4 = 29$$
$$(4-a)^2 = 25$$
$$4-a = \pm 5$$
$$a = -1 \text{ or } 9$$

6 **NAPLAN A**

Calculate the values of p if the distance between $(p, 3)$ and $(1, p)$ are exactly 10 units apart.

$$\sqrt{(1-p)^2+(p-3)^2} = 10$$
$$(1-p)^2+(p-3)^2 = 100$$
$$1-2p+p^2+p^2-6p+9 = 100$$
$$2p^2-8p-90 = 0$$
$$p^2-4p-45 = 0$$
$$(p-9)(p+5) = 0$$
$$p = 9 \text{ or } -5$$

7 Points A, B and C lie on a straight line.
Point A has coordinates $(-4, 1)$
Point B has coordinates $(7, -5)$
Point C is such that $AB : BC = 4 : 7$.
Work out the length of line segment AC. Give your answer to 1 decimal place.

$$AB^2 = \left(7-(-4)\right)^2+(-5-1)^2 = 157$$
$$AB = \sqrt{157}$$
$$BC = \frac{7}{4}AB$$
$$= \frac{7}{4}\sqrt{157}$$

$$AC = AB + BC$$
$$= AB + \frac{7}{4}AB$$
$$= \frac{11}{4}AB$$
$$= \frac{11}{4}\sqrt{157}$$
$$= 34.4$$